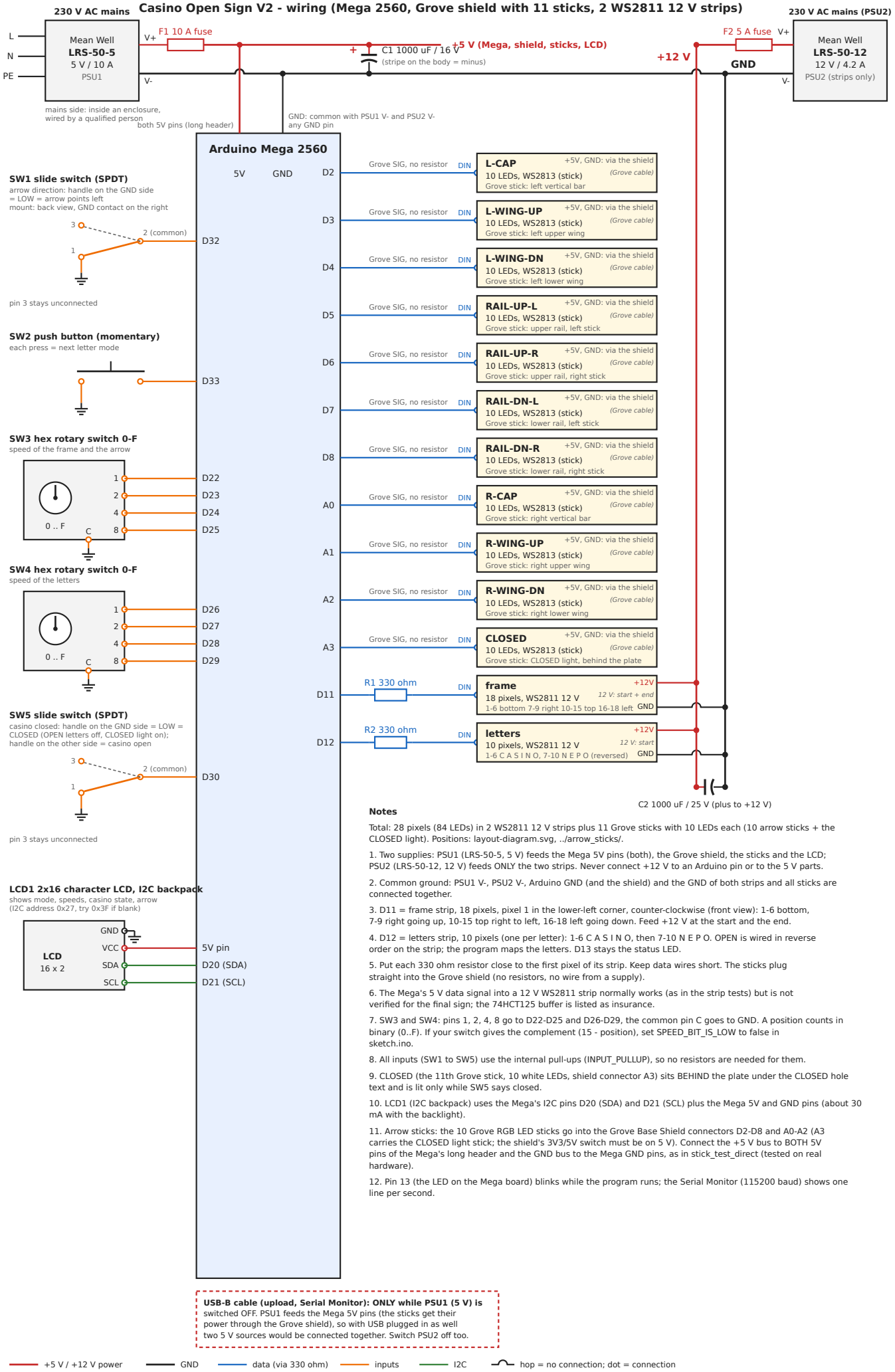

Casino Open Sign V2 - wiring (Mega 2560, Grove shield with 11 sticks, 2 WS2811 12 V strips)
230 V AC mains
L
N
PE
Mean Well
LRS-50-5
5 V / 10 A
PSU1
V+
V-
F1 10 A fuse
mains side: inside an enclosure, wired by a qualified person
both 5V pins (long header)
+
C1 1000 uF / 16 V
(stripe on the body = minus)
+5 V (Mega, shield, sticks, LCD)
GND: common with PSU1 V- and PSU2 V- any GND pin
+12 V
GND
F2 5 A fuse
230 V AC mains (PSU2)
Mean Well
LRS-50-12
12 V / 4.2 A
PSU2 (strips only)
Arduino Mega 2560
5V
GND
SW1 slide switch (SPDT)
arrow direction: handle on the GND side = LOW = arrow points left
mount: back view, GND contact on the right
3
2 (common)
1
D32
pin 3 stays unconnected
SW2 push button (momentary)
each press = next letter mode
D33
SW3 hex rotary switch 0-F
speed of the frame and the arrow
0 .. F
C
1
2
4
8
D22
D23
D24
D25
SW4 hex rotary switch 0-F
speed of the letters
D26
D27
D28
D29
SW5 slide switch (SPDT)
casino closed: handle on the GND side = LOW = CLOSED (OPEN letters off, CLOSED light on); handle on the other side = casino open
D30
pin 3 stays unconnected
LCD1 2x16 character LCD, I2C backpack
shows mode, speeds, casino state, arrow
(I2C address 0x27, try 0x3F if blank)
LCD
16 x 2
GND
VCC
SDA
SCL
5V pin
D20 (SDA)
D21 (SCL)
D2
D3
D4
D5
D6
D7
D8
A0
A1
A2
A3
D11
D12
Grove SIG, no resistor
DIN
L-CAP
10 LEDs, WS2813 (stick)
Grove stick: left vertical bar
L-WING-UP
Grove stick: left upper wing
L-WING-DN
Grove stick: left lower wing
RAIL-UP-L
Grove stick: upper rail, left stick
RAIL-UP-R
Grove stick: upper rail, right stick
RAIL-DN-L
Grove stick: lower rail, left stick
RAIL-DN-R
Grove stick: lower rail, right stick
R-CAP
Grove stick: right vertical bar
R-WING-UP
Grove stick: right upper wing
R-WING-DN
Grove stick: right lower wing
CLOSED
Grove stick: CLOSED light, behind the plate
+5V, GND: via the shield (Grove cable)
R1 330 ohm
R2 330 ohm
frame
18 pixels, WS2811 12 V
1-6 bottom 7-9 right 10-15 top 16-18 left
+12V
12 V: start + end
GND
letters
10 pixels, WS2811 12 V
1-6 C A S I N O, 7-10 N E P O (reversed)
12 V: start
C2 1000 uF / 25 V (plus to +12 V)
Notes
Total: 28 pixels (84 LEDs) in 2 WS2811 12 V strips plus 11 Grove sticks with 10 LEDs each (10 arrow sticks + the CLOSED light). Positions: layout-diagram.svg, ../arrow_sticks/.
1. Two supplies: PSU1 (LRS-50-5, 5 V) feeds the Mega 5V pins (both), the Grove shield, the sticks and the LCD; PSU2 (LRS-50-12, 12 V) feeds ONLY the two strips. Never connect +12 V to an Arduino pin or to the 5 V parts.
2. Common ground: PSU1 V-, PSU2 V-, Arduino GND (and the shield) and the GND of both strips and all sticks are connected together.
3. D11 = frame strip, 18 pixels, pixel 1 in the lower-left corner, counter-clockwise (front view): 1-6 bottom, 7-9 right going up, 10-15 top right to left, 16-18 left going down. Feed +12 V at the start and the end.
4. D12 = letters strip, 10 pixels (one per letter): 1-6 C A S I N O, then 7-10 N E P O. OPEN is wired in reverse order on the strip; the program maps the letters. D13 stays the status LED.
5. Put each 330 ohm resistor close to the first pixel of its strip. Keep data wires short. The sticks plug straight into the Grove shield (no resistors, no wire from a supply).
6. The Mega's 5 V data signal into a 12 V WS2811 strip normally works (as in the strip tests) but is not verified for the final sign; the 74HCT125 buffer is listed as insurance.
7. SW3 and SW4: pins 1, 2, 4, 8 go to D22-D25 and D26-D29, the common pin C goes to GND. A position counts in binary (0..F). If your switch gives the complement (15 - position), set SPEED_BIT_IS_LOW to false in sketch.ino.
8. All inputs (SW1 to SW5) use the internal pull-ups (INPUT_PULLUP), so no resistors are needed for them.
9. CLOSED (the 11th Grove stick, 10 white LEDs, shield connector A3) sits BEHIND the plate under the CLOSED hole text and is lit only while SW5 says closed.
10. LCD1 (I2C backpack) uses the Mega's I2C pins D20 (SDA) and D21 (SCL) plus the Mega 5V and GND pins (about 30 mA with the backlight).
11. Arrow sticks: the 10 Grove RGB LED sticks go into the Grove Base Shield connectors D2-D8 and A0-A2 (A3 carries the CLOSED light stick; the shield's 3V3/5V switch must be on 5 V). Connect the +5 V bus to BOTH 5V pins of the Mega's long header and the GND bus to the Mega GND pins, as in stick_test_direct (tested on real hardware).
12. Pin 13 (the LED on the Mega board) blinks while the program runs; the Serial Monitor (115200 baud) shows one line per second.
USB-B cable (upload, Serial Monitor): ONLY while PSU1 (5 V) is switched OFF. PSU1 feeds the Mega 5V pins (the sticks get their power through the Grove shield), so with USB plugged in as well two 5 V sources would be connected together. Switch PSU2 off too.
+5 V / +12 V power
GND
data (via 330 ohm)
inputs
I2C
hop = no connection; dot = connection